Experiment 7

Aim: Apply K-Means and Hierarchical Clustering on sample datasets.

Problem statement: In modern business environments, customer retention is a critical challenge for service-based companies such as telecom providers. Customers often discontinue services due to pricing, service quality, or competition.
The objective of this experiment is to build an Artificial Neural Network (ANN) model that can analyze customer behavior data and predict whether a customer will churn (leave the service) or stay. This helps companies take proactive measures to improve customer retention and reduce revenue loss.

Dataset Link: Customer Churn Prediction Dataset
https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Dataset Description

The dataset contains telecom customer details and their service usage patterns.

- **Features include:**
  - Gender
  - SeniorCitizen
  - Tenure
  - MonthlyCharges
  - TotalCharges
  - Contract Type
  - Internet Service
  - etc.
- **Target Variable:**
  - **Churn** (Yes = 1, No = 0)
- **Dataset Characteristics:**
  - Mixed dataset (categorical + numerical)
  - Binary classification problem
  - Real-world customer behavior dataset

Mathematical Formulation of the Algorithm

Artificial Neural Network (ANN)

$$a = \sigma(Wx + b)$$

- x = input features
- W = weights
- b = bias
- sigma = activation function

Sigmoid Activation Function

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

- Outputs probability between 0 and 1

Algorithm Limitations

- Requires large dataset for best performance
- Computationally expensive
- Difficult to interpret (black-box model)
- Prone to overfitting if not tuned properly

Methodology:

1. Data Collection
The dataset was imported from Kaggle and loaded into the environment.
2. Data Preprocessing
- Removed unnecessary column: customerID
- Converted target variable:
  - Yes → 1
  - No → 0
- Encoded categorical variables using Label Encoding
- Handled missing values by removing null entries

3. Feature Selection
- **Input Features:** All variables except Churn
- **Target Variable:** Churn

4. Feature Scaling
StandardScaler was applied to normalize all features for better ANN performance.
5. Train-Test Split
- 80% training data
- 20% testing data
- Random state used for reproducibility

6. Model Architecture
- Input Layer: 16 neurons (ReLU activation)

- Hidden Layer: 8 neurons (ReLU activation)
- Output Layer: 1 neuron (Sigmoid activation)

7. Model Training

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Epochs: 20
- Batch Size: 32

8. Evaluation

Model performance was evaluated using:

- Accuracy
- Loss curve
- Precision, Recall, F1-score

9. Hyperparameter Tuning

Different configurations of:

- Number of layers
- Neurons per layer
- Epochs
- Batch size

were tested to improve model performance.

Performance Analysis

- **Evaluation Metrics Used:**
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Loss curve

Observations

- ANN captures complex patterns in customer data
- Accuracy improves with proper tuning
- Loss decreases over epochs indicating learning

Hyperparameter Tuning

- Tuned parameters:
  - Number of layers
  - Number of neurons
  - Epochs
  - Batch size

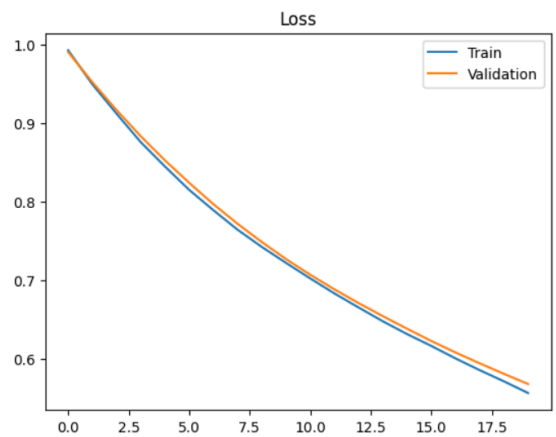
Impact

- Improved accuracy and model performance
- Reduced overfitting
- Better generalization

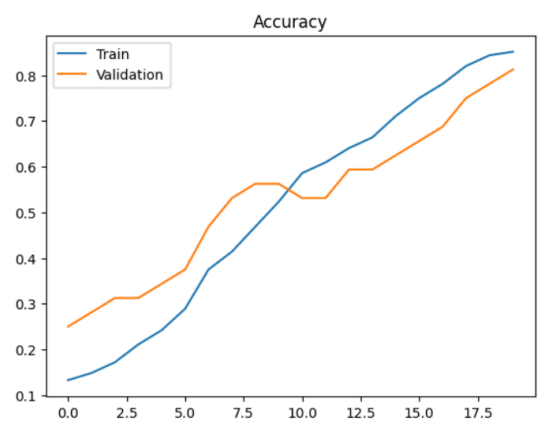
Result:
Classification report

	precision	recall	f1-score	support
0	0.92	0.89	0.90	37
1	0.00	0.00	0.00	3
accuracy			0.82	40
macro avg	0.46	0.45	0.45	40
weighted avg	0.85	0.82	0.84	40

Loss validation



Accuracy



SDG 8: Improves business growth by reducing customer churn
SDG 9: Uses ANN for advanced predictive analytics
SDG 12: Supports efficient service usage and customer retention